EE 533 Information Theory

Week 03

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Kullback-Leibler Divergence

- Let p and q be PMFs on the countable set $\mathcal{X}$, then

$$D(p\|q) := \begin{cases} \sum_x p(x) \log \frac{p(x)}{q(x)} & \text{if } p \prec q \\ \infty & \text{else} \end{cases}$$

  where $p \prec q$ iff $\nexists x$ satisfying both $p(x) > 0$ and $q(x) = 0$.
- Let X and Y be discrete random variables with the PMFs p_X and p_Y, then

$$D(p_X\|p_Y) = \mathbf{E}\left[\log \frac{p_X(X)}{p_Y(X)}\right]$$

- KL Div. is also called the relative entropy or the Kullback-Leibler distance, see [CT06].
- $D(p_X\|p_Y) \neq D(p_Y\|p_X)$:
  If X is either one or zero with equal probability and Y is always one, then

$$D(p_X\|p_Y) = \infty \qquad \text{because } p_X \not\prec p_Y.$$
$$D(p_Y\|p_X) = \log 2$$

Theorem (Non-negativity of the KL divergence)

$$D(p_X \| p_Y) \geq 0 \qquad \forall X, Y$$

Furthermore, $D(p_X \| p_Y) = 0$ iff $p_X = p_Y$.

Proof:

$$\begin{aligned}
D(p_X \| p_Y) &= \mathbf{E}\left[-\log \frac{p_Y(X)}{p_X(X)}\right] \\
&\overset{(a)}{\geq} -\log \mathbf{E}\left[\frac{p_Y(X)}{p_X(X)}\right] && \text{by Jensen's ineq. and the convexity of } \log \tfrac{1}{\tau} \text{ in } \tau \\
&= -\log \mathbf{E}\left[\mathbb{1}_{\{p_X(Y)>0\}}\right] && \text{because } \mathbf{E}[f(X)] = \mathbf{E}\left[f(Y)\frac{p_X(Y)}{p_Y(Y)}\mathbb{1}_{\{p_X(Y)>0\}}\right] \text{ for } p_X \prec p_Y \quad (\star) \\
&\overset{(b)}{\geq} 0 && \text{by } \mathbf{E}\left[\mathbb{1}_{\{p_X(Y)>0\}}\right] = \mathbf{P}(p_X(Y)>0) \text{ and } \mathbf{P}(p_X(Y)>0) \leq 1
\end{aligned}$$

The inequality (b) is strict unless $\mathbf{P}(p_X(Y) > 0) = 1$ and the inequality (a) is strict unless $\frac{p_Y(X)}{p_X(X)} = \gamma$ with probability one. Thus $D(p_X \| p_Y) = 0$ iff $\frac{p_Y(X)}{p_X(X)} = 1$, i.e. $p_X = p_Y$ $\square$

Remark

- *For $D(p_X \| p_Y) = 0$, we only need the equality of the distribution not the random variables!*
- *$(\star)$ is called the measure change argument and $\mathbb{1}_{\{p_X(Y)>0\}}$ can be dropped (**H.W.**).*

Theorem (Joint Convexity of the Kullback-Leibler divergence)

$$D(p_{X_\beta}\|p_{Y_\beta}) \le \beta D(p_{X_1}\|p_{Y_1})+(1-\beta)D(p_{X_0}\|p_{Y_0}) \qquad \forall \beta \in [0,1],\ \forall X_0, X_1, Y_0, Y_1,$$

where $p_{X_\beta}=\beta p_{X_1}+(1-\beta)p_{X_0}$ *and* $p_{Y_\beta}=\beta p_{Y_1}+(1-\beta)p_{Y_0}$. *Furthermore, inequality is strict unless* $p_{X_1}p_{Y_0} = p_{X_0}p_{Y_1}$.

Proof.

$$\begin{aligned}
\beta D(p_{X_1}\|p_{Y_1})+(1-\beta)D(p_{X_0}\|p_{Y_0}) &= \beta \mathbf{E}\left[\log \frac{p_{X_1}(X_1)}{p_{Y_1}(X_1)}\right] + (1-\beta)\mathbf{E}\left[\log \frac{p_{X_0}(X_0)}{p_{Y_0}(X_0)}\right] \\
&= \beta \mathbf{E}\left[\frac{p_{X_1}(X_\beta)}{p_{X_\beta}(X_\beta)} \log \frac{p_{X_1}(X_\beta)}{p_{Y_1}(X_\beta)}\right] + (1-\beta)\mathbf{E}\left[\frac{p_{X_0}(X_\beta)}{p_{X_\beta}(X_\beta)} \log \frac{p_{X_0}(X_\beta)}{p_{Y_0}(X_\beta)}\right] \\
&= \mathbf{E}\left[\beta \frac{p_{X_1}(X_\beta)}{p_{X_\beta}(X_\beta)}\left(-\log \frac{p_{Y_1}(X_\beta)}{p_{X_1}(X_\beta)}\right) + (1-\beta)\frac{p_{X_0}(X_\beta)}{p_{X_\beta}(X_\beta)}\left(-\log \frac{p_{Y_0}(X_\beta)}{p_{X_0}(X_\beta)}\right)\right] \\
&\overset{(a)}{\ge} \mathbf{E}\left[-\log\left(\beta \frac{p_{X_1}(X_\beta)}{p_{X_\beta}(X_\beta)}\frac{p_{Y_1}(X_\beta)}{p_{X_1}(X_\beta)} + (1-\beta)\frac{p_{X_0}(X_\beta)}{p_{X_\beta}(X_\beta)}\frac{p_{Y_0}(X_\beta)}{p_{X_0}(X_\beta)}\right)\right] \\
&= D(p_{X_\beta}\|p_{Y_\beta})
\end{aligned}$$

(a) is by Jensen's ineq. and the convexity of $\log \frac{1}{\tau}$. (a) is strict unless $\frac{p_{X_1}(X_\beta)}{p_{Y_1}(X_\beta)} = \frac{p_{X_0}(X_\beta)}{p_{Y_0}(X_\beta)}$. □

Theorem (Monotonicity)

$$D(p_{\mathsf{X}}\|p_{\mathsf{Y}}) \geq D\left(p_{\widetilde{\mathsf{X}}}\middle\|p_{\widetilde{\mathsf{Y}}}\right) \qquad \forall f,\ \textit{where } \widetilde{\mathsf{X}}{:=}f(\mathsf{X}) \textit{ and } \widetilde{\mathsf{Y}}{:=}f(\mathsf{Y}).$$

The inequality is strict unless $\frac{p_{\mathsf{Y}}(x)}{p_{\mathsf{X}}(x)}=\frac{p_{\mathsf{Y}}(z)}{p_{\mathsf{X}}(z)}$ *holds whenever* $f(x)=f(z)$ *and* $p_{\mathsf{X}}(z)\,p_{\mathsf{X}}(x)>0$.

Proof.

$$\begin{aligned}
D(p_{\mathsf{X}}\|p_{\mathsf{Y}}) &= \mathbf{E}\left[-\log\frac{p_{\mathsf{Y}}(\mathsf{X})}{p_{\mathsf{X}}(\mathsf{X})}\right] \\
&= \mathbf{E}\left[\mathbf{E}\left[-\log\frac{p_{\mathsf{Y}}(\mathsf{X})}{p_{\mathsf{X}}(\mathsf{X})}\middle|\widetilde{\mathsf{X}}\right]\right] && \text{by the law of iterated expectations.} \\
&\overset{(a)}{\geq} \mathbf{E}\left[-\log\mathbf{E}\left[\frac{p_{\mathsf{Y}}(\mathsf{X})}{p_{\mathsf{X}}(\mathsf{X})}\middle|\widetilde{\mathsf{X}}\right]\right] && \text{by Jensen's ineq and the convexity of } \log\tfrac{1}{\tau} \\
&= \mathbf{E}\left[-\log\frac{p_{\widetilde{\mathsf{Y}}}(\widetilde{\mathsf{X}})}{p_{\widetilde{\mathsf{X}}}(\widetilde{\mathsf{X}})}\right] && \text{by } p_{\mathsf{X}|\widetilde{\mathsf{X}}}(x|z)=\frac{p_{\mathsf{X}}(x)\mathbb{1}_{\{z=f(x)\}}}{p_{\widetilde{\mathsf{X}}}(z)} \ \&\ p_{\widetilde{\mathsf{Y}}}(z)=\sum\nolimits_x p_{\mathsf{Y}}(x)\mathbb{1}_{\{z=f(x)\}} \\
&= D\left(p_{\widetilde{\mathsf{X}}}\middle\|p_{\widetilde{\mathsf{Y}}}\right)
\end{aligned}$$

□

- Recall that $H(\mathsf{X}) \geq H(f(\mathsf{X}))$ with equality iff f is one-to-one (i.e. injective).
- **H.W.** Monotonicity theorem implies the non-negativity theorem and convexity theorem.

Definition (Joint and Conditional Kullback-Leibler Divergence)

Let X and Y be $\mathcal{X}$ valued r.v.'s and Z and T be $\mathcal{Z}$ valued r.v.'s, then

$$D(p_{\mathsf{X},\mathsf{Z}}\|p_{\mathsf{Y},\mathsf{T}}) := \mathbf{E}\left[\log \frac{p_{\mathsf{X},\mathsf{Z}}(\mathsf{X},\mathsf{Z})}{p_{\mathsf{Y},\mathsf{T}}(\mathsf{X},\mathsf{Z})}\right] \qquad \text{(Joint Kullback-Leibler Divergence)}$$

$$D\left(p_{\mathsf{X}|\mathsf{Z}}\middle\|p_{\mathsf{Y}|\mathsf{T}}\middle|p_{\mathsf{Z}}\right) := \mathbf{E}\left[\log \frac{p_{\mathsf{X}|\mathsf{Z}}(\mathsf{X}|\mathsf{Z})}{p_{\mathsf{Y}|\mathsf{T}}(\mathsf{X}|\mathsf{Z})}\right] \qquad \text{(Conditional Kullback-Leibler Divergence)}$$

Theorem (The Chain Rule for Kullback-Leibler Divergence)

Let X and Y be $\mathcal{X}$ valued r.v.'s and Z and T be $\mathcal{Z}$ valued r.v.'s, then

$$\begin{aligned} D(p_{\mathsf{X},\mathsf{Z}}\|p_{\mathsf{Y},\mathsf{T}}) &= D(p_{\mathsf{Z}}\|p_{\mathsf{T}}) + D\left(p_{\mathsf{X}|\mathsf{Z}}\middle\|p_{\mathsf{Y}|\mathsf{T}}\middle|p_{\mathsf{Z}}\right) \\ &= D(p_{\mathsf{X}}\|p_{\mathsf{Y}}) + D\left(p_{\mathsf{Z}|\mathsf{X}}\middle\|p_{\mathsf{T}|\mathsf{Y}}\middle|p_{\mathsf{X}}\right) \end{aligned}$$

Proof.

$$\begin{aligned} D(p_{\mathsf{X},\mathsf{Z}}\|p_{\mathsf{Y},\mathsf{T}}) &= \mathbf{E}\left[\log \frac{p_{\mathsf{Z}}(\mathsf{Z})}{p_{\mathsf{T}}(\mathsf{Z})} + \log \frac{p_{\mathsf{X}|\mathsf{Z}}(\mathsf{X}|\mathsf{Z})}{p_{\mathsf{Y}|\mathsf{T}}(\mathsf{X}|\mathsf{Z})}\right] && \text{chain rule for probabilites} \\ &= D(p_{\mathsf{X}}\|p_{\mathsf{Y}}) + D\left(p_{\mathsf{Z}|\mathsf{X}}\middle\|p_{\mathsf{T}|\mathsf{Y}}\middle|p_{\mathsf{X}}\right) \end{aligned}$$

□

Remark

- *Usually $q_{\mathsf{X},\mathsf{Z}}$ is used instead of $p_{\mathsf{Y},\mathsf{T}}$ and $q_{\mathsf{X}|\mathsf{Z}}$ is used instead of $p_{\mathsf{Y}|\mathsf{T}}$ to denote the PMFs in the second argument of the KLD.*

Chain Rule (General)

Theorem

If X_i and Y_i are $\mathcal{X}_i$ valued discrete random variables for each $i \in \{1, \ldots, n\}$, then

$$D\left(p_{X_1^n} \middle\| p_{Y_1^n}\right) = D(p_{X_1} \| p_{Y_1}) + \sum_{i=2}^{n} D\left(p_{X_i|X_1^{i-1}} \middle\| p_{Y_i|Y_1^{i-1}} \middle| p_{X_1^{i-1}}\right)$$

Proof.

$$\begin{aligned}
D\left(p_{X_1^n} \middle\| p_{Y_1^n}\right) &= \mathbf{E}\left[\ln \frac{p_{X_1^n}(X_1^n)}{p_{Y_1^n}(X_1^n)}\right] \\
&= \mathbf{E}\left[\ln\left(\frac{p_{X_1}(X_1)}{p_{Y_1}(X_1)} \prod_{i=2}^{n} \frac{p_{X_i|X_1^{i-1}}(X_i|X_1^{i-1})}{p_{Y_i|Y_1^{i-1}}(X_i|X_1^{i-1})}\right)\right] \\
&= \mathbf{E}\left[\ln \frac{p_{X_1}(X_1)}{p_{Y_1}(X_1)}\right] + \sum_{i=2}^{n} \mathbf{E}\left[\ln \frac{p_{X_i|X_1^{i-1}}(X_i|X_1^{i-1})}{p_{Y_i|Y_1^{i-1}}(X_i|X_1^{i-1})}\right] \\
&= D(p_{X_1} \| p_{Y_1}) + \sum_{i=2}^{n} D\left(p_{X_i|X_1^{i-1}} \middle\| p_{Y_i|Y_1^{i-1}} \middle| p_{X_1^{i-1}}\right)
\end{aligned}$$

□

Implications of The Chain Rule for KL Divergence

$$D(p_{X,Z}\|p_{Y,T}) = D(p_X\|p_Y) + D\big(p_{Z|X}\big\|p_{T|Y}\big|p_X\big) \qquad (*)$$
$$= D(p_Z\|p_T) + D\big(p_{X|Z}\big\|p_{Y|T}\big|p_Z\big) \qquad (**)$$

Theorem

(a) $D(p_{X,Z}\|p_{Y,T}) \geq D(p_X\|p_Y)$

(b) *If* $p_{Z|X} = p_{T|Y}$ *then* $D(p_X\|p_Y) \geq D(p_Z\|p_T)$ *"also called DPI for KLD"*

(c) *If* $p_X = p_Y$ *then* $D\big(p_{Z|X}\big\|p_{T|Y}\big|p_X\big) \geq D(p_Z\|p_T)$

Proof.

(a) Follows from (*) because $D\big(p_{Z|X}\big\|p_{T|Y}\big|p_X\big) \geq 0$.

(b) $p_{Z|X} = p_{T|Y}$ implies $D\big(p_{Z|X}\big\|p_{T|Y}\big|p_X\big) = 0$ and hence $D(p_X\|p_Y) = D(p_{X,Z}\|p_{Y,T})$ by (*). Thus (b) follows from (**) because $D\big(p_{X|Z}\big\|p_{Y|T}\big|p_Z\big) \geq 0$.

(c) $p_X = p_Y$ implies $D(p_X\|p_Y) = 0$ and hence $D\big(p_{Z|X}\big\|p_{T|Y}\big|p_X\big) = D(p_{X,Z}\|p_{Y,T})$ by (*). Thus (c) follows from (**) because $D\big(p_{X|Z}\big\|p_{Y|T}\big|p_Z\big) \geq 0$.

Chain Rule for KL Divergence and Product Probabilities

Theorem

If $\mathsf{X}_\imath$ *and* $\mathsf{Y}_\imath$ *are* $\mathcal{X}_\imath$ *valued discrete random variables for each* $\imath \in \{1, \ldots, n\}$*, then*

$$D\left(p_{\mathsf{X}_1^n} \middle\| p_{\mathsf{Y}_1} \cdots p_{\mathsf{Y}_n}\right) = D\left(p_{\mathsf{X}_1^n} \middle\| p_{\mathsf{X}_1} \cdots p_{\mathsf{X}_n}\right) + \sum_{\imath=1}^{n} D(p_{\mathsf{X}_\imath} \| p_{\mathsf{Y}_\imath}).$$

Proof.

$$\begin{aligned} D\left(p_{\mathsf{X}_1^n} \middle\| p_{\mathsf{Y}_1} \cdots p_{\mathsf{Y}_n}\right) &= \mathbf{E}\left[\log \frac{p_{\mathsf{X}_1^n}(\mathsf{X}_1^n)}{p_{\mathsf{X}_1}(\mathsf{X}_1)\cdots p_{\mathsf{X}_n}(\mathsf{X}_n)} \frac{p_{\mathsf{X}_1}(\mathsf{X}_1)\cdots p_{\mathsf{X}_n}(\mathsf{X}_n)}{p_{\mathsf{Y}_1}(\mathsf{X}_1)\cdots p_{\mathsf{Y}_n}(\mathsf{X}_n)}\right] \\ &= \mathbf{E}\left[\log \frac{p_{\mathsf{X}_1^n}(\mathsf{X}_1^n)}{p_{\mathsf{X}_1}(\mathsf{X}_1)\cdots p_{\mathsf{X}_n}(\mathsf{X}_n)}\right] + \sum_{\imath=1}^{n} \mathbf{E}\left[\log \frac{p_{\mathsf{X}_\imath}(\mathsf{X}_\imath)}{p_{\mathsf{Y}_\imath}(\mathsf{X}_\imath)}\right] \end{aligned}$$

□

Corollary

If $\mathsf{X}_\imath$ *and* $\mathsf{Y}_\imath$ *are* $\mathcal{X}_\imath$ *valued discrete random variables for each* $\imath \in \{1, \ldots, n\}$*, then*

$$D\left(p_{\mathsf{X}_1^n} \middle\| p_{\mathsf{Y}_1} \cdots p_{\mathsf{Y}_n}\right) \geq \sum_{\imath=1}^{n} D(p_{\mathsf{X}_\imath} \| p_{\mathsf{Y}_\imath})$$

Furthermore, the equality holds iff $p_{\mathsf{X}_1^n} = p_{\mathsf{X}_1} \cdots p_{\mathsf{X}_n}$.

Rényi Divergence

Is the Kulback-Leibler divergence the only divergence satisfying the additivity property? No!

- Let $\alpha \in \mathbb{R}_+ \setminus \{1\}$ and p, q be PMFs on the countable set $\mathcal{X}$, then
$$D_\alpha(p\|q) := \frac{1}{\alpha-1} \log \sum_x p(x) \left[\frac{q(x)}{p(x)}\right]^{1-\alpha}.$$
- Let $\alpha \in \mathbb{R}_+ \setminus \{1\}$ and X, Y be discrete random variables with the PMFs p_X and p_Y, then
$$D_\alpha(p_X\|p_Y) = \frac{1}{\alpha-1} \log \mathbf{E}\left[\left(\frac{p_Y(X)}{p_X(X)}\right)^{1-\alpha}\right]$$
- Rényi provided an axiomatic characterization [Rén61]. Near exhaustive study is provided in [EH14]. Some of the key observations are:
$$\lim_{\alpha\to 1} D_\alpha(p\|q) = D(p\|q)$$
$$D_\alpha(p\|q) \geq 0 \qquad \text{equality hods iff } q = p$$
$$D_\alpha(p_{X_1}\cdots p_{X_n}\|p_{Y_1}\cdots p_{Y_n}) = \sum_{i=1}^n D_\alpha(p_{X_i}\|p_{Y_i})$$

Entropy and KL Divergence (Rényi Entropy and Rényi Divergence)

Lemma

Let $\mathcal{X}$ be a finite set and U be random variable with uniform distribution on $\mathcal{X}$. Then any $\mathcal{X}$ valued random variable X satisfies

$$D(p_{\mathsf{X}} \| p_{\mathsf{U}}) = \log |\mathcal{X}| - H(\mathsf{X})$$
$$D_\alpha(p_{\mathsf{X}} \| p_{\mathsf{U}}) = \log |\mathcal{X}| - H_\alpha(\mathsf{X})$$

Proof.

$$\begin{aligned}
D(p_{\mathsf{X}} \| p_{\mathsf{U}}) &= \mathbf{E}\left[\log \frac{p_{\mathsf{X}}(\mathsf{X})}{p_{\mathsf{U}}(\mathsf{X})}\right] \\
&= \log |\mathcal{X}| - \mathbf{E}\left[\log \frac{1}{p_{\mathsf{X}}(\mathsf{X})}\right] && \text{because } p_{\mathsf{U}}(x) = 1/|\mathcal{X}| \text{ for all } x \\
D_\alpha(p_{\mathsf{X}} \| p_{\mathsf{U}}) &= \frac{1}{\alpha-1} \log \mathbf{E}\left[\left(\frac{p_{\mathsf{U}}(\mathsf{X})}{p_{\mathsf{X}}(\mathsf{X})}\right)^{1-\alpha}\right] \\
&= \log |\mathcal{X}| - \frac{1}{1-\alpha} \log \mathbf{E}\left[(p_{\mathsf{X}}(\mathsf{X}))^{\alpha-1}\right] && \text{because } p_{\mathsf{U}}(x) = 1/|\mathcal{X}| \text{ for all } x
\end{aligned}$$

□

Variational Characterization of Rényi Entropy/Divergence with KLD

Lemma

Let $\alpha \in \mathbb{R}_+ \setminus \{1\}$ and X, Y be discrete r.v.'s s.t. $H_\alpha(X) < \infty$ and $D_\alpha(p_X \| p_Y) < \infty$. Then

$$(1-\alpha)H_\alpha(X) = \sup_{p_Z} (1-\alpha)H(Z) - \alpha D(p_Z \| p_X)$$

$$(1-\alpha)D_\alpha(p_X \| p_Y) = \inf_{p_Z} (1-\alpha)D(p_Z \| p_Y) + \alpha D(p_Z \| p_X)$$

Proof.

$$\begin{aligned}
(1-\alpha)H(Z) - \alpha D(p_Z \| p_X) &= \mathbf{E}\left[(1-\alpha)\log\frac{1}{p_Z(Z)} - \alpha \log\frac{p_Z(Z)}{p_X(Z)}\right] \\
&= \mathbf{E}\left[\log\frac{(p_X(Z))^\alpha}{p_Z(Z)}\right] \\
&= (1-\alpha)H_\alpha(X) - D(p_Z \| p_{X_\alpha}) \qquad \text{for } p_{X_\alpha}(\tau) := \frac{(p_X(\tau))^\alpha}{2^{(1-\alpha)H_\alpha(X)}} \quad \forall \tau \\
(1-\alpha)D(p_Z \| p_Y) + \alpha D(p_Z \| p_X) &= \mathbf{E}\left[(1-\alpha)\log\frac{p_Z(Z)}{p_Y(Z)} + \alpha \log\frac{p_Z(Z)}{p_X(Z)}\right] \\
&= \mathbf{E}\left[\log\frac{p_Z(Z)}{(p_Y(Z))^{(1-\alpha)}(p_X(Z))^\alpha}\right] \\
&= (1-\alpha)D_\alpha(p_X \| p_Y) + D(p_Z \| p_{X_\alpha}) \qquad \text{for } p_{X_\alpha}(\tau) := \frac{(p_X(\tau))^\alpha (p_Y(\tau))^{1-\alpha}}{2^{(\alpha-1)D_\alpha(p_X \| p_Y)}}
\end{aligned}$$

$\square$

Variational Characterizations of Mutual Information with KL Divergence

Lemma

Let X, Y *be discrete random variables s.t.* $I(\mathrm{X};\mathrm{Y}) < \infty$. *Then*

$$\begin{aligned} I(\mathrm{X};\mathrm{Y}) &= D(p_{\mathrm{X},\mathrm{Y}} \| p_{\mathrm{X}} p_{\mathrm{Y}}) \\ &= \inf_{p_{\mathrm{Z}}} D(p_{\mathrm{X},\mathrm{Y}} \| p_{\mathrm{X}} p_{\mathrm{Z}}) \\ &= \inf_{p_{\mathrm{T}}, p_{\mathrm{Z}}} D(p_{\mathrm{X},\mathrm{Y}} \| p_{\mathrm{T}} p_{\mathrm{Z}}) \end{aligned}$$

Proof.

$$\begin{aligned} D(p_{\mathrm{X},\mathrm{Y}} \| p_{\mathrm{X}} p_{\mathrm{Y}}) &= \mathbf{E}\left[\log \frac{p_{\mathrm{X},\mathrm{Y}}(\mathrm{X},\mathrm{Y})}{p_{\mathrm{X}}(\mathrm{X}) p_{\mathrm{Y}}(\mathrm{Y})}\right] & &= H(\mathrm{X}) + H(\mathrm{Y}) - H(\mathrm{X},\mathrm{Y}) \\ &= I(\mathrm{X};\mathrm{Y}) \\ D(p_{\mathrm{X},\mathrm{Y}} \| p_{\mathrm{X}} p_{\mathrm{Z}}) &= \mathbf{E}\left[\log \frac{p_{\mathrm{X},\mathrm{Y}}(\mathrm{X},\mathrm{Y})}{p_{\mathrm{X}}(\mathrm{X}) p_{\mathrm{Z}}(\mathrm{Y})}\right] & &= \mathbf{E}\left[\log \frac{p_{\mathrm{X},\mathrm{Y}}(\mathrm{X},\mathrm{Y})}{p_{\mathrm{X}}(\mathrm{X}) p_{\mathrm{Y}}(\mathrm{Y})} + \log \frac{p_{\mathrm{Y}}(\mathrm{Y})}{p_{\mathrm{Z}}(\mathrm{Y})}\right] \\ &= I(\mathrm{X};\mathrm{Y}) + D(p_{\mathrm{Y}} \| p_{\mathrm{Z}}) \\ D(p_{\mathrm{X},\mathrm{Y}} \| p_{\mathrm{T}} p_{\mathrm{Z}}) &= \mathbf{E}\left[\log \frac{p_{\mathrm{X},\mathrm{Y}}(\mathrm{X},\mathrm{Y})}{p_{\mathrm{T}}(\mathrm{X}) p_{\mathrm{Z}}(\mathrm{Y})}\right] & &= \mathbf{E}\left[\log \frac{p_{\mathrm{X},\mathrm{Y}}(\mathrm{X},\mathrm{Y})}{p_{\mathrm{X}}(\mathrm{X}) p_{\mathrm{Y}}(\mathrm{Y})} + \log \frac{p_{\mathrm{X}}(\mathrm{X})}{p_{\mathrm{T}}(\mathrm{X})} + \log \frac{p_{\mathrm{Y}}(\mathrm{Y})}{p_{\mathrm{Z}}(\mathrm{Y})}\right] \\ &= I(\mathrm{X};\mathrm{Y}) + D(p_{\mathrm{X}} \| p_{\mathrm{T}}) + D(p_{\mathrm{Y}} \| p_{\mathrm{Z}}) \end{aligned}$$

□

A more carefull analysis shows finiteness hypothesis is not necessary.

Mutual Information and KL Divergence

Theorem

(a) *$I(\mathsf{X};\mathsf{Y})$ is convex in $p_{\mathsf{Y}|\mathsf{X}}$ for a given p_{X}.*

(b) *$I(\mathsf{X};\mathsf{Y})$ is concave in p_{X} for a given $p_{\mathsf{Y}|\mathsf{X}}$.*

Proof.

(a)
$$\begin{aligned}
\beta I(\mathsf{X};\mathsf{Y}_1) + (1-\beta) I(\mathsf{X};\mathsf{Y}_0) &= \beta D\big(p_{\mathsf{X}} p_{\mathsf{Y}_1|\mathsf{X}} \big\| p_{\mathsf{X}} p_{\mathsf{Y}_1}\big) + (1-\beta) D\big(p_{\mathsf{X}} p_{\mathsf{Y}_0|\mathsf{X}} \big\| p_{\mathsf{X}} p_{\mathsf{Y}_0}\big) \\
&\overset{(a)}{\geq} D\big(\beta p_{\mathsf{X}} p_{\mathsf{Y}_1|\mathsf{X}} + (1-\beta) p_{\mathsf{X}} p_{\mathsf{Y}_0|\mathsf{X}} \big\| \beta p_{\mathsf{X}} p_{\mathsf{Y}_1} + (1-\beta) p_{\mathsf{X}} p_{\mathsf{Y}_0}\big) \\
&= D\Big(p_{\mathsf{X}} p_{\mathsf{Y}_\beta|\mathsf{X}} \Big\| p_{\mathsf{X}} p_{\mathsf{Y}_\beta}\Big) \\
&= I(\mathsf{X};\mathsf{Y}_\beta)
\end{aligned}$$

where (a) follows from the joint convexity of KLD and $p_{\mathsf{Y}_\beta|\mathsf{X}} = \beta p_{\mathsf{Y}_1|\mathsf{X}} + (1-\beta) p_{\mathsf{Y}_0|\mathsf{X}}$.

(b) Recall that $I(\mathsf{X};\mathsf{Y}) = \inf_{p_{\mathsf{Z}}} D(p_{\mathsf{X},\mathsf{Y}} \| p_{\mathsf{X}} p_{\mathsf{Z}})$ and $D(p_{\mathsf{X},\mathsf{Y}} \| p_{\mathsf{X}} p_{\mathsf{Z}}) = D\big(p_{\mathsf{Y}|\mathsf{X}} \big\| p_{\mathsf{Z}} \big| p_{\mathsf{X}}\big)$ is linear and hence concave in p_{X} for a given $p_{\mathsf{Y}|\mathsf{X}}$ and p_{Z}. Then $I(\mathsf{X};\mathsf{Y})$ is concave in p_{X} for a given $p_{\mathsf{Y}|\mathsf{X}}$ because pointwise infimum of any collection of concave functions is concave, i.e. because $f(\tau) := \inf_\gamma f_\gamma(\tau)$ is concave in τ provided that $f_\gamma(\tau)$ is concave in τ for all γ.

Theorem

If $X_1, X_2, \ldots$ *are independent identically distributed (i.i.d.) with* p_X, $Y_1, Y_2, \ldots$ *are i.i.d. with* p_Y, *and* $D(p_X \| p_Y) < \infty$, *then* $\frac{1}{n} \log \frac{p_{X_1^n}(X_1^n)}{p_{Y_1^n}(X_1^n)} \xrightarrow{\mathbf{P}} D(p_X \| p_Y)$

Proof.

$$\frac{1}{n} \log \frac{p_{X_1^n}(X_1^n)}{p_{Y_1^n}(X_1^n)} = \frac{1}{n} \sum_{i=1}^{n} Z_i \qquad \text{where } Z_i := \log \frac{p_X(X_i)}{p_Y(X_i)}$$

$\mathbf{E}[Z_i] \neq \mathbf{E}[|Z_i|]$ because Z_i can take negative values.

$$\begin{aligned}
\mathbf{E}\left[|Z_i| \mathbb{1}_{\{Z_i<0\}}\right] &= \mathbf{E}\left[\log \frac{p_Y(X_i)}{p_X(X_i)} \mathbb{1}_{\{Z_i<0\}}\right] \\
&= \mathbf{E}\left[\frac{p_X(Y_i)}{p_Y(Y_i)} \log \frac{p_Y(Y_i)}{p_X(Y_i)} \mathbb{1}_{\{Z_i<0\}}\right] \qquad \textbf{H.W. } \text{Why?} \\
&\leq \sup_{z \in [0,1]} z \log \frac{1}{z} \\
&= \frac{\log e}{e} \qquad \textbf{H.W. } \text{Prove } \sup_{z \in [0,1]} z \log \frac{1}{z} = \frac{\log e}{e}
\end{aligned}$$

Then $\mathbf{E}[|Z_i|] \leq \mathbf{E}[Z_i] + 2\frac{\log e}{e} = D(p_X \| p_Y) + 2\frac{\log e}{e}$ and theorem follows from the W.L.L.N. □

Corollary

If $(X_1, Y_1), (X_2, Y_2), \ldots$ *are i.i.d.* $\sim p_{X,Y}$ *and* $I(X;Y) < \infty$ *then* $\frac{1}{n} \log \frac{p_{X_1^n, Y_1^n}(X_1^n, Y_1^n)}{p_{X_1^n}(X_1^n) p_{Y_1^n}(Y_1^n)} \xrightarrow{\mathbf{P}} I(X;Y)$

Definition

$$\mathcal{A}_\epsilon^{(n)} := \left\{ x_1^n : 2^{n(D(p_X\|p_Y)-\epsilon)} \le \frac{p_{X_1^n}(x_1^n)}{p_{Y_1^n}(x_1^n)} \le 2^{n(D(p_X\|p_Y)+\epsilon)} \right\}$$

Theorem

If $X_1, X_2, \ldots$ *are i.i.d.* $\sim p_X$, $Y_1, Y_2, \ldots$ *are i.i.d.* $\sim p_Y$, *and* $D(p_X\|p_Y) < \infty$, *then for any* $\delta > 0$

(a) $\mathbf{P}\left(X_1^n \in \mathcal{A}_\epsilon^{(n)}\right) > 1 - \delta$ *for sufficiently large* n.

(b) $\mathbf{P}\left(Y_1^n \in \mathcal{B} \cap \mathcal{A}_\epsilon^{(n)}\right) \le 2^{-n(D(p_X\|p_Y)-\epsilon)} \mathbf{P}\left(X_1^n \in \mathcal{B} \cap \mathcal{A}_\epsilon^{(n)}\right)$

(c) $\mathbf{P}(Y_1^n \in \mathcal{B}) \ge 2^{-n(D(p_X\|p_Y)+\epsilon)} (\mathbf{P}(X_1^n \in \mathcal{B}) - \delta)$ *for sufficiently large* n.

Proof.

(a) W.L.L.N.

(b) $\mathbf{P}(Y_1^n = x_1^n) \le 2^{-n(D(p_X\|p_Y)-\epsilon)} \mathbf{P}(X_1^n = x_1^n)$ for all $x_1^n \in \mathcal{A}_\epsilon^{(n)}$

(c) $\mathbf{P}(X_1^n \in \mathcal{B}) \le \mathbf{P}\left(X_1^n \notin \mathcal{A}_\epsilon^{(n)}\right) + \mathbf{P}\left(X_1^n \in \mathcal{B} \cap \mathcal{A}_\epsilon^{(n)}\right) \le \delta + \mathbf{P}\left(X_1^n \in \mathcal{B} \cap \mathcal{A}_\epsilon^{(n)}\right)$.

$\mathbf{P}\left(X_1^n \in \mathcal{B} \cap \mathcal{A}_\epsilon^{(n)}\right) \le 2^{n(D(p_X\|p_Y)+\epsilon)} \mathbf{P}\left(Y_1^n \in \mathcal{B} \cap \mathcal{A}_\epsilon^{(n)}\right) \le 2^{n(D(p_X\|p_Y)+\epsilon)} \mathbf{P}(Y_1^n \in \mathcal{B})$.

□

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